

Solving *La Ghigliottina*

with a Typed PPMI Word Graph

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Introduction and Motivations

La Ghigliottina (final round of *L'Eredità*): five clue words, find the one word that connects to **all five**.

The link is **not** semantic similarity — it is **collocational and cultural**: the answer forms a fixed multi-word expression, compound, or idiom with each clue.

legno · forte · continua · malattia · veloce $\Rightarrow$ **cassa**

cassa di legno · cassaforte · cassa continua · cassa malattia · cassa

Distributional models (word2vec/fastText) are a poor structural fit — the signal is syntagmatic, not paradigmatic.

Related Work

The EVALITA **NLP4FUN** shared task framed *La Ghigliottina* as a computational language game. The seminal player **OTTHO** (Basile et al. 2009) solved it by **knowledge infusion** — retrieval / co-occurrence over a dictionary, proverb lists, and Wikipedia — and **beat learned-embedding methods**, the outcome we reproduce.

Our contribution is not a new model but a controlled ablation showing **why**:

- On 30k entities, knowledge-graph embeddings are dominated by **hubness**.
- Every recall-broadening extension trades top-1 accuracy for rank-tail recall.
- The signal is **syntagmatic**, so word2vec / fastText are a poor structural fit.

Proposed Approach

1. **Typed, weighted word graph** from PAISÀ. Edge *types* are MWE-bearing patterns — MOD (noun–adj), BINOMIAL (X e Y), DI (X di Y), DA (X da Y). Edge *weights* are **pattern-conditioned PPMI** (PMI within each relation’s own pairs).
2. **Curated IDIOM edges** from Italian *polirematiche* + proverbs ($\approx 18.5\text{k}$ pairs). They supply the clue→answer links running text can’t — added at a constant weight.
3. **Flat PPMI-sum solver**. Per clue take the strongest edge to a candidate (max over relations/directions), then sum across the five clues. Highest sum wins.

No learned parameters. One tuned scalar (the idiom weight).

Optional --describe pass: a local **Qwen3-1.7B** explains the answer, one sentence per clue — post-hoc, plays no part in solving.

Main result

Full PAISÀ graph (600k triples / 32k entities), EVALITA 2018 test (100 games).

CONFIGURATION	FOUND	MRR	ACC@1	ACC@5	ACC@10	ACC@100
flat-PPMI(sum), PAISÀ only	60	0.288	0.240	0.330	0.380	0.600
+ IDIOM (w=5)	67	0.394	0.340	0.460	0.520	0.670
IDIOM alone (no corpus)	43	0.206	0.160	0.240	0.270	0.430

Curated idiom edges are the **single largest lever** in the project: acc@1 0.24 → **0.34** (+42% rel), MRR 0.29 → 0.39. Weight is a wide plateau (w = 4.5–6), not a knife-edge fit.

Description quality

The optional --describe explanations (Qwen3-1.7B) scored vs. the gold desc field (100 test games), generated for the solver's **own** top-1 prediction:

BLEU	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum	BERTScore
1.70	0.248	0.035	0.177	0.170	0.658

BLEU on the sacreBLEU 0–100 scale; the rest 0–1.

Low n-gram overlap — free-form Italian, generated for the **predicted** word (acc@1 0.34) — but **BERTScore 0.66** shows the explanations are semantically on-topic. A usability layer, not a scored task.

Explored Extensions and Negative Results

Every extension improves recall / MRR / $\text{acc}@_{10,100}$ — **none improves $\text{acc}@1$.**

Models / scorers

- KGE (ComplEx, RotatE) — $\text{acc}@1$ 0.04 (hubness)
- Personalised PageRank — $\text{acc}@1$ 0.05
- Coverage / logsum scorers
- Learned re-ranker (10 features)

More data / more edges

- Wikipedia, itWaC joins (register miss)
- OpenSubtitles (helps recall, not top-1)
- COMPOUND, SIM (fastText) edges
- Wiktionary MWEs, ConceptNet

8 independent experiments · all confirm the same ceiling

Knowledge-graph embeddings — decisive loss

The project's **original** thesis: learned embeddings could score *unobserved* clue→answer links a graph baseline can't.

METHOD	FOUND	MRR	ACC@1
flat-PPMI(sum)	60	0.288	0.240
KGE ComplEx (300 ep)	21	0.023	0.010
KGE RotatE (150 ep)	37	0.068	0.040

Over 30k entities, link prediction is dominated by **hubness** — frequent adjectives / prepositions absorb probability mass, so the model smears its score instead of pinning the collocate (known partners rank #66–170).

Additional corpora — register beats volume

Extra corpora joined into the PPMI scan (PAISÀ-defined candidates):

ADDED CORPUS	+TOKENS	ACC@1	ACC@100	VERDICT
— (PAISÀ only)	—	0.240	0.600	baseline
Wikipedia	+650M	0.240	0.610	recall only
OpenSubtitles 12M	+85M	0.270	0.630	helps
OpenSubtitles full	+766M	0.250	0.660	recall ↑, top-1 ↓
itWaC (web)	+93M	0.220	0.600	hurts

Dialogue (85M tokens) beats Wikipedia (which **doubled** the graph). But nothing beats the curated idiom edges — the **purest** register match.

The recall-miss anatomy — why the ceiling exists

Gold-rank anatomy of the 100 test games:

RANK #1	RANK 2–10	RANK 11–100	MISSING (>100)
34	18	17	31

- The 31 misses **exist as graph nodes** but lack the clue→answer **edges** (98/100 gold already in vocab — coverage is not the problem).
- A learned re-ranker lifts **train** acc@1 but **does not transfer** — the ceiling is not a ranking artefact.
- SIM / ConceptNet leave train found essentially unchanged — wrong **kind** of edge (paradigmatic / too sparse), not the missing syntagmatic collocations.

Conclusions and Limitations

A **minimal** system — typed pattern-PPMI over PAISÀ + curated idiom edges + flat sum — reaches **acc@1 0.34 / MRR 0.39** on EVALITA 2018 NLP4FUN, beating KGE, PageRank, and every corpus/edge extension.

The lesson: the task is bottlenecked by **register-matched recall**, not model sophistication. The one lever that ever moved top-1 was the idiom register.

Next step: a register-matched, associative **phrase** resource — not a smarter ranker or more generic text (both dead ends).

Thank you for your attention!